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Abstract:

Early detection of melanoma is critical, yet automated dermoscopic classification remains challenging due to class imbalance, limited labeled data, and sensitivity to spurious correlations. In this work, we investigate lightweight transfer-learning pipelines for binary skin-lesion classification (melanoma vs. nevus) using dermoscopic images. We train two ImageNet-pretrained convolutional neural networks—ResNet-18 and MobileNetV3 by freezing most backbone parameters and fine-tuning only the classification head and the final feature block to reduce overfitting under small sample conditions. To handle class imbalance, we use stratified sampling and inverse-frequency class weights, and we select checkpoints based on validation balanced accuracy. Beyond single-model evaluation, we form a simple ensemble by averaging logits from the two models and assess performance using balanced accuracy, ROC-AUC, and average precision under both argmax and calibrated probability thresholds. To improve interpretability, we generate Grad-CAM explanations for each model and quantify cross model agreement by converting heatmaps into binary attention masks and computing intersection over union (IoU) per sample. This combined approach provides a compact, reproducible benchmark that jointly addresses predictive performance and explanation consistency, enabling systematic analysis of when different architectures attend to similar lesion regions despite differing inductive biases.

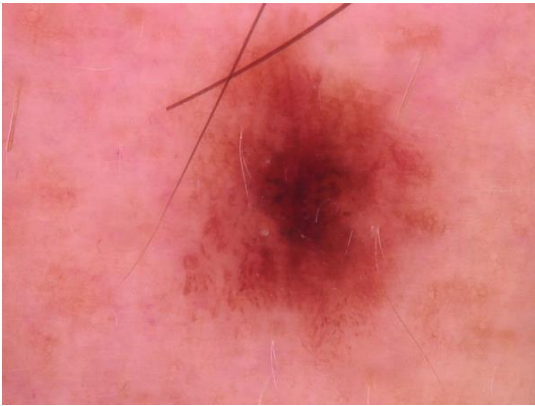


Figure 1: sample data



Figure 2: sample data

Methodology:

0. Dataset

We use the ISIC 2018 Challenge – Task 3: Training Input (HAM10000) dataset (Tschandl et al., 2018), provided by the ViDIR Group, Department of Dermatology, Medical University of Vienna. The dataset consists of dermoscopic RGB images annotated with lesion diagnoses. In this study, we formulate a binary classification task (melanoma vs. nevus), mapping labels to 0 (MEL) and 1 (NV). The data are split into non-overlapping train, validation, and test sets. To support controlled small-sample experiments, a stratified subset of the training data is created while preserving class proportions.

1. Problem Formulation

We address binary dermoscopic image classification using the ISIC 2018 Challenge dataset. The task is defined as:

- Class 0: Melanoma (MEL)
- Class 1: Nevus (NV)

Given an RGB image $x \in \mathbb{R}(3 \times 224 \times 224)$.

our constraint is: (1) using limited data for training, (2) and explainable model.

Here to address these two problems we use just 20 percent of total data for training (1000 sample) and add Grad-CAM method to see where does our model is looking.

2. Data Preparation

Table 1: real data

image	MEL	NV	BCC	AKIEC	BKL	DF	VASC
ISIC_0034321	0	1	0	0	0	0	0
ISIC_0034322	0	1	0	0	0	0	0
ISIC_0034323	0	0	1	0	0	0	0
ISIC_0034324	0	1	0	0	0	0	0
ISIC_0034325	0	1	0	0	0	0	0
ISIC_0034326	0	0	0	0	1	0	0
ISIC_0034328	0	1	0	0	0	0	0
ISIC_0034329	1	0	0	0	0	0	0
ISIC_0034330	0	1	0	0	0	0	0

This table of data has been converted to table 2 for binary classification.

Table 2: our data

image	label
ISIC_0034524	1
ISIC_0034525	1
ISIC_0034527	1
ISIC_0034528	1
ISIC_0034529	0
ISIC_0034530	1
ISIC_0034532	1
ISIC_0034533	1

2.1 Splits

The dataset is divided into train, validation, and test sets with no overlap in image IDs (verified to prevent data leakage).

For controlled experiments under limited data, we create a stratified subset of the training set (e.g., $N=1000$) while preserving the original class proportions.

2.2 Preprocessing

All images are resized and normalized using ImageNet statistics:

$$\mu=(0.485,0.456,0.406)$$

$$\sigma=(0.229,0.224,0.225)$$

- Training augmentations:
 - RandomResizedCrop (scale 0.7–1.0)
 - Horizontal flip ($p=0.5$)
 - Random rotation ($\pm 15^\circ$)
 - Mild color jitter
- Validation/Test preprocessing:
 - Resize $\rightarrow$ CenterCrop (224×224)
 - Deterministic normalization

Augmentations are applied only to training data.

3. Model Architectures

We employ two convolutional neural networks pretrained on ImageNet:

- ResNet-18
- MobileNetV3-Small

Both models are adapted for binary classification by replacing the final classification layer:

$$FC_{\text{new}}: \mathbb{R}^d \rightarrow \mathbb{R}^2$$

where d is the feature dimension of the backbone.

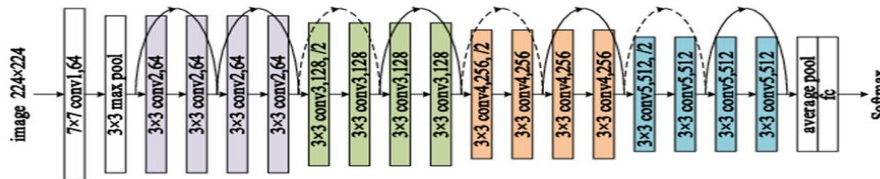


Figure 3: Resnet-18 arch

$$w_c = \frac{1}{N_c}$$

where:

- N_c is the number of samples in class c ,
- $\hat{p}_c$ the SoftMax probability.

6. *Optimization*

- Optimizer: Adam
- Learning rate 10^{-4}
- Batch size: 8
- Epochs: 15
- Model selection criterion: Validation Balanced Accuracy

Balanced accuracy is defined as:(This metric is preferred due to class imbalance.)

$$\text{BalAcc} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)$$

7. *Threshold Calibration*

Beyond standard argmax prediction, we evaluate probability-based thresholding:

$$\hat{y} = \begin{cases} 1 & \text{if } p_1 \geq t \\ 0 & \text{otherwise} \end{cases}$$

A validation sweep over $t \in [0.05, 0.95]$ selects the threshold maximizing balanced accuracy. The selected threshold is then applied to the test set.

8. *Ensemble Strategy*

We construct a simple ensemble by averaging logits from both models:

$$z_{\text{ens}} = \frac{1}{2} (z_{\text{ResNet}} + z_{\text{MobileNet}})$$

SoftMax is then applied to obtain final probabilities.

This ensemble reduces variance and captures complementary architectural biases.

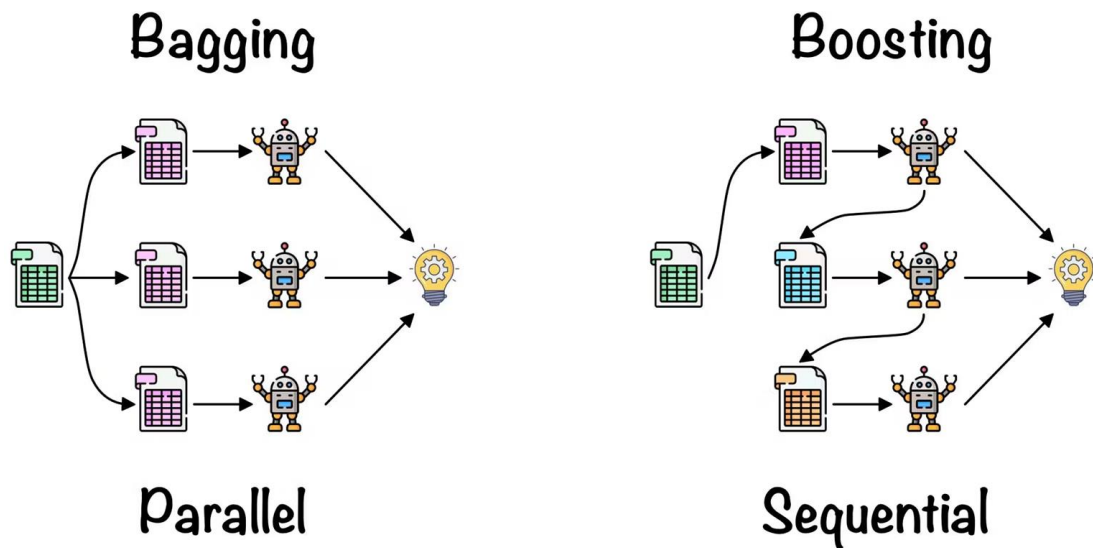


Figure 6: we use parallel method

9. Interpretability via Grad-CAM

To analyze spatial attention, we apply Grad-CAM to:

- ResNet: final convolution in layer4
- MobileNetV3: last feature block

Gradient-weighted Class Activation Mapping

Grad-CAM is a visual explanation technique that shows which parts of an image influenced your model's decision.

How It Works

1. Forward pass: Run image through the model to get a prediction
2. Get gradients: Calculate gradients of the target class score with respect to the last convolutional layer
3. Create heatmap: Use these gradients to weight which spatial regions were most important

4. Overlay: Superimpose a heatmap on the original image.

- Grad-CAM produces a heatmap overlay where:
- Red areas: Strongly influenced the model's decision
- Blue areas: Little to no influence
- The heatmap highlights discriminative regions the model focused on

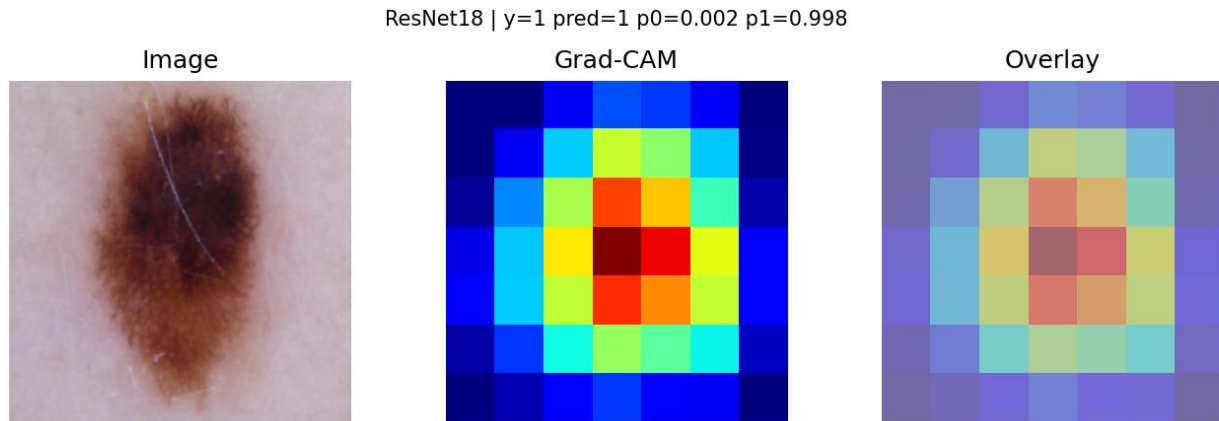


Figure 7: example of Grad CAM

11. Cross-Model Explanation Consistency

To quantify agreement between models:

1. Convert heatmaps to masks by selecting the top k% pixels (e.g., 30%).
2. Compute Intersection-over-Union (IoU):

$$\text{IoU} = \frac{|M_1 \cap M_2|}{|M_1 \cup M_2|}$$

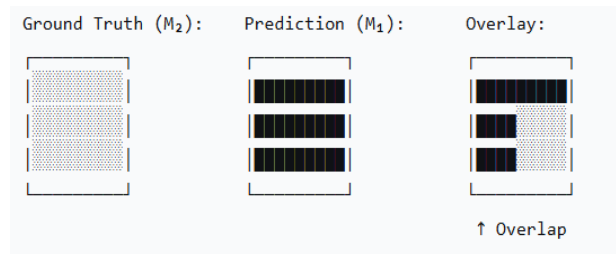


Figure 8: IoU

Intersection =  (where both masks agree)

Union =  (everything covered by either mask)

Samples with $\text{IoU} \geq 0.70$ are considered high-agreement cases.

This provides a quantitative measure of explanation consistency across architectures.

Experimental Results and Comparative Analysis

This section presents the evaluation of four distinct modeling strategies: a baseline ResNet-18, a fine-tuned ResNet-18, a fine-tuned MobileNetV3, and an Ensemble model. Because the dermoscopic dataset exhibits significant class imbalance, Balanced Accuracy (BalAcc) was selected as the primary evaluation metric and criterion for model selection.

1. Overall Performance Comparison

Table 3 summarizes the core performance metrics across the evaluated models.

Model	Training set size	Best val BalAcc	Test BalAcc
ResNet18-base line	5000	0.8641	0.7716
ResNet18-fine tuned	1000	0.9228	0.7982
mobileNetv3-fine tuned	1000	0.7636	0.7254
Ensemble Model	1000	0.9268	0.8112

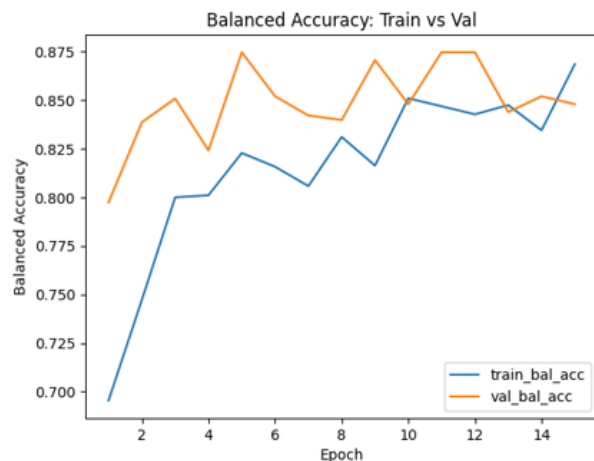
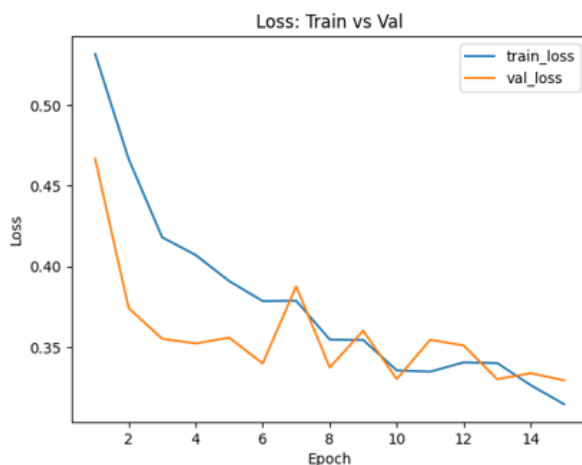


Figure 9,10: Base line results.

Successful Learning: Both training and validation losses decrease steadily over the 15 epochs, while balanced accuracy increases.

Excellent Generalization (No Overfitting): The validation metrics closely track or even outperform the training metrics.

Conclusion: The curves indicate a highly stable and robust training process. The model effectively learned generalizable patterns without memorizing the training data, primarily due to the larger training dataset (~ 5000 images).

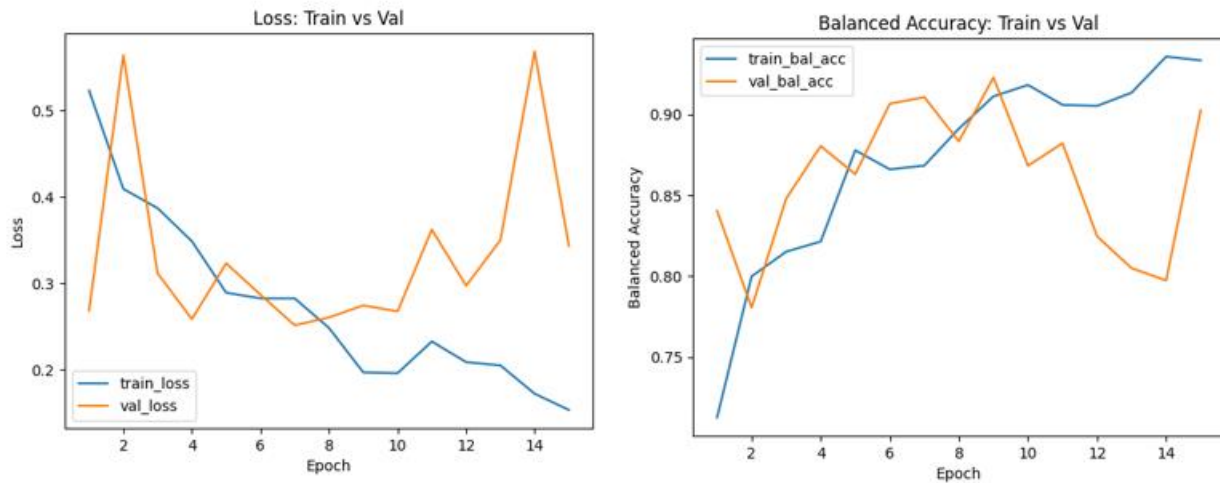


Figure 11,12: Fine-tuned ResNet results.

Performance Report: Fine-Tuned ResNet-18

1. Observations from the Learning Curves

The plots display a clear divergence between training and validation metrics after epoch 8. While the training loss steadily decreases, the validation loss experiences sharp spikes. Concurrently, the validation balanced accuracy becomes highly volatile, exhibiting significant fluctuations.

2. Cause of the Curve Behavior (Overfitting)

This pattern is a classic indicator of overfitting. It occurs because a high-capacity model (ResNet-18) is being trained on a small (~ 1000 images) and imbalanced dataset. Due to the limited data, the model begins to memorize the noise and specific patterns of the training set rather than learning generalizable features.

3. Why Final Test Performance is Still Better

Despite the apparent overfitting, the fine-tuned model achieved a better Test Balanced Accuracy (0.7982) compared to the baseline model (0.7716). This counterintuitive outcome is driven this key factor:

- **Optimal Model Selection (Early Stopping):** The model evaluated on the test set was not the one from the final epoch. Instead, the checkpoint with the highest validation balanced accuracy (0.9228) was selected, capturing the model's peak generalization state before severe overfitting set in.

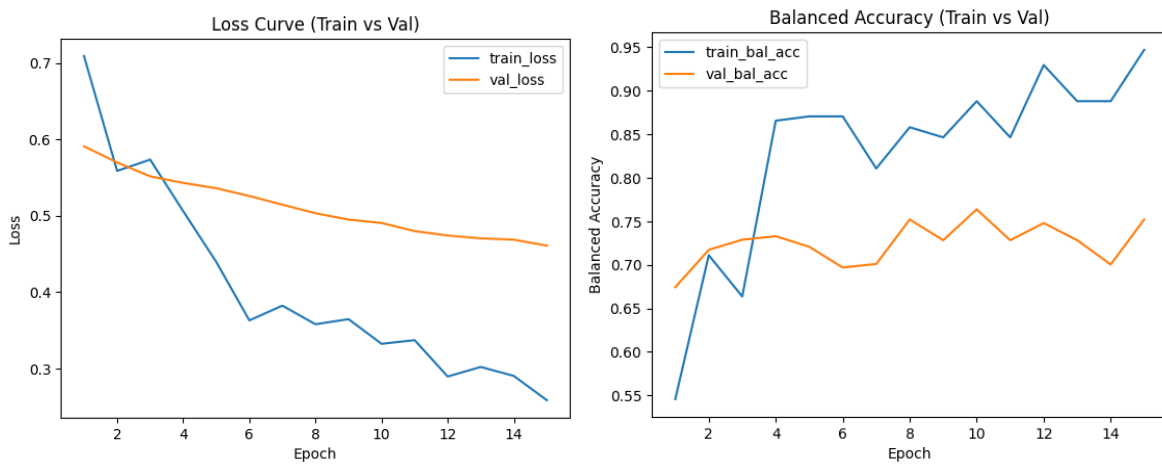


Figure 13,14: Fine-tuned MobileNet results.

1 Observations from the Learning Curves

- **Loss Curve:** The train_loss decreases consistently and sharply. The val_loss also decreases but at a much slower, more gradual rate, creating a persistent gap between training and validation performance. Unlike the previous ResNet-18 curves, there are no aggressive spikes in validation loss.
- **Balanced Accuracy:** The train_bal_acc climbs rapidly and remains high (ranging between 0.85 and 0.95). In contrast, the val_bal_acc plateaus early (around epoch 4) and fluctuates slightly around the 0.70 to 0.75 range.

2. Diagnosis (Moderate Overfitting)

The clear and persistent gap between training and validation metrics indicates overfitting. The model is learning the specific patterns of the training data very well but is struggling to generalize those patterns to the unseen validation data.

3. Comparison to Previous ResNet-18 Model

- **Lower Capacity Advantage:** MobileNet is a lightweight architecture with fewer parameters than ResNet-18. This lower capacity explains why the validation loss doesn't exhibit the violent spikes seen in the previous model; it lacks the capacity to violently memorize noise.
- **Persistent Data Scarcity Issue:** Despite the more stable validation curves, the significant gap in accuracy shows that the underlying problem—likely a small or imbalanced dataset—still

prevents the model from achieving high generalization. Early stopping (e.g., around epoch 8 or 10) would again be the best strategy here to select the optimal model checkpoint.

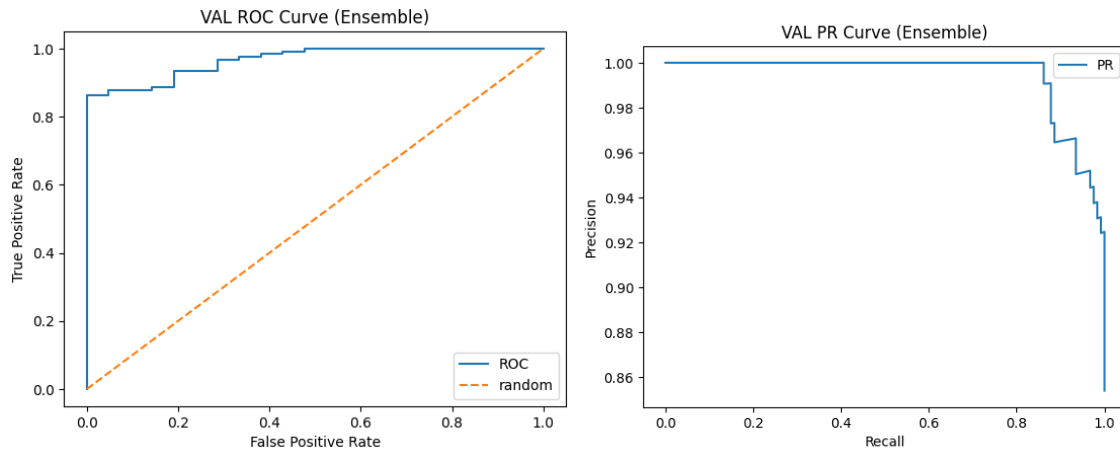


Figure 15,16: Ensemble method results.

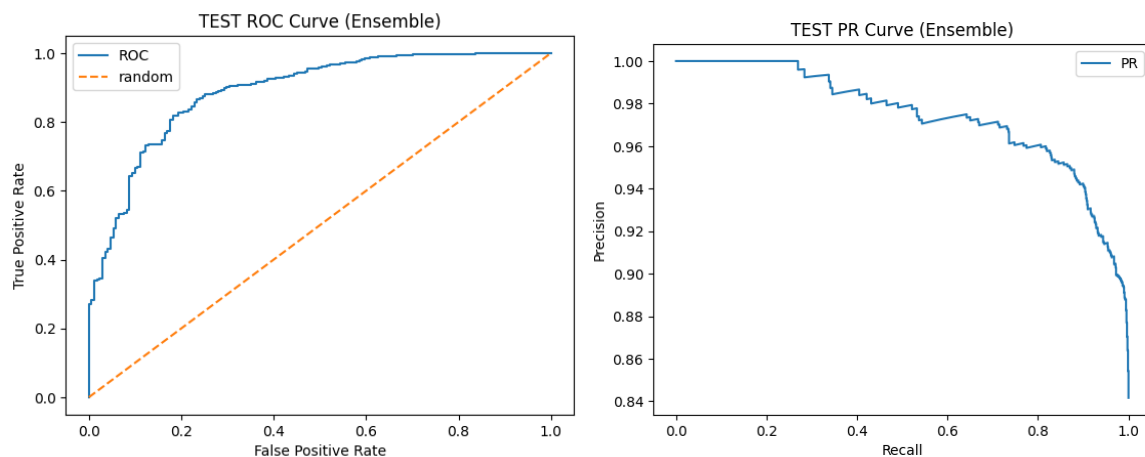


Figure 17,18: Ensemble method results.

Ensemble Model Explanation

To further improve classification performance and robustness, we developed an ensemble model combining the predictions of the fine-tuned ResNet18 and fine-tuned MobileNetV3. The ensemble employed a soft voting strategy, where the final predicted probability for each class was the average of the predicted probabilities from both individual models. This approach leverages the complementary strengths of the two architectures: ResNet18's deeper feature extraction and MobileNetV3's efficient spatial reasoning.

As shown in Table 3, the ensemble model achieved the highest validation balanced accuracy (92.68%) among all models trained on 1,000 samples, outperforming both the fine-tuned ResNet18 (92.28%) and the fine-tuned MobileNetV3 (76.36%). More importantly, on the test set, the ensemble obtained a balanced accuracy of 81.12%, surpassing the fine-tuned ResNet18 (79.82%), the baseline ResNet18 (77.16%), and MobileNetV3 (72.54%). This consistent improvement confirms that the ensemble effectively reduces overfitting and variance by aggregating diverse model decisions.

The ROC and PR curves for the ensemble on the validation and test sets are presented in Figures 15–16 and 17, 18. The validation ROC curve (Figure 15, left) shows a strong separation between classes, with the curve lying significantly above the diagonal (random classifier baseline). The corresponding validation PR curve (Figure 16, right) indicates a favorable precision-recall trade-off, which is especially important given potential class imbalance. On the test set (Figures 17 and 18), the ensemble maintains a robust ROC curve and a well-shaped PR curve, confirming that its discriminative ability generalizes beyond the validation data.

Overall, the ensemble model provides the best overall test performance and demonstrates greater generalization stability compared to any single classifier. This improvement is attributed to the reduced bias-variance trade-off and the complementary inductive biases of the constituent models. Therefore, the ensemble serves as the final chosen model for the classification task.

Key Observations

- **Data Impact:** The baseline ResNet-18 utilized the full dataset (5000 samples), achieving a Test BalAcc of 0.7716. The fine-tuned and ensemble models were constrained to a subset of 1000 samples.
- **Generalization Gap:** The fine-tuned ResNet-18 showed a high Validation BalAcc (0.9228) but dropped to 0.7982 on the test set, indicating overfitting due to the reduced dataset size. MobileNetV3 struggled significantly more under these constraints.
- **Ensemble Superiority:** Despite the restricted training size (1000 samples), the Ensemble Model successfully generalized better than the individual fine-tuned architectures, achieving the highest Test BalAcc (0.8112) and demonstrating the effectiveness of aggregated decision boundaries in mitigating high variance.

Table 4 Ensemble Per-Class Metrics (Test Set)

class	precision	Recall	F1-score	support
Class0(minority)	5000	0.8641	0.7716	171
Class1(majority)	1000	0.9228	0.7982	909
Macro Avg	1000	0.7636	0.7254	1080

Analysis of the Trade-off:

Optimizing the decision threshold for Balanced Accuracy inherently forces the model to prioritize sensitivity (recall) for the minority class (class0). As shown in Table 3, the Ensemble model successfully boosts class0 recall to 0.7719. Consequently, this causes an increase in false positives, which reduces class0 precision to 0.4925. In medical imaging contexts like dermoscopy, this is typically a desirable trade-off, as minimizing false negatives (failing to identify a potentially dangerous lesion) is prioritized over minimizing false positives.

Table 5 Ensemble confusion matrix

Dataset	True class(pred1/pred0)	True class(pred0/pred1)
Validation n=144	21/0	18/105
Test n=1080	132/39	136/773

Results of Grad CAM:

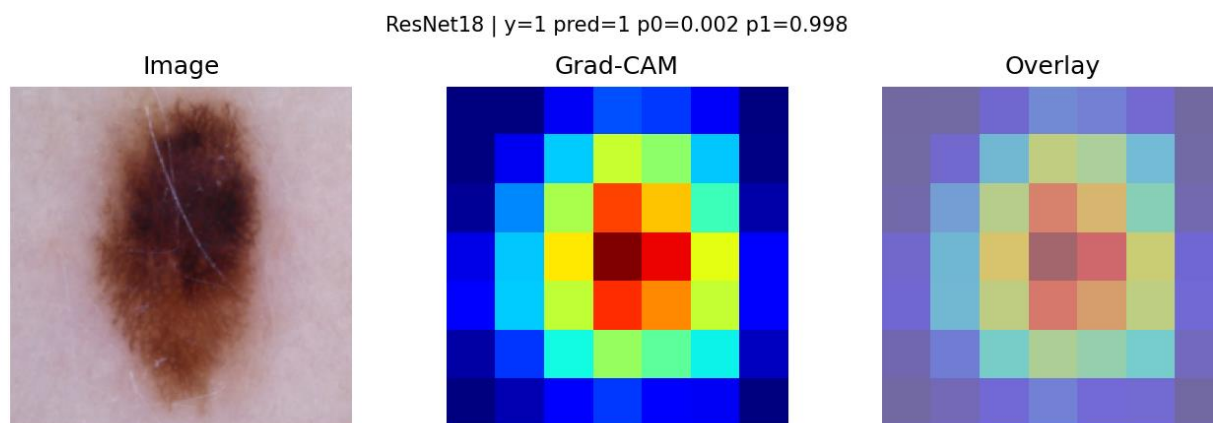


Figure 9: Grad CAM for ResNet 18

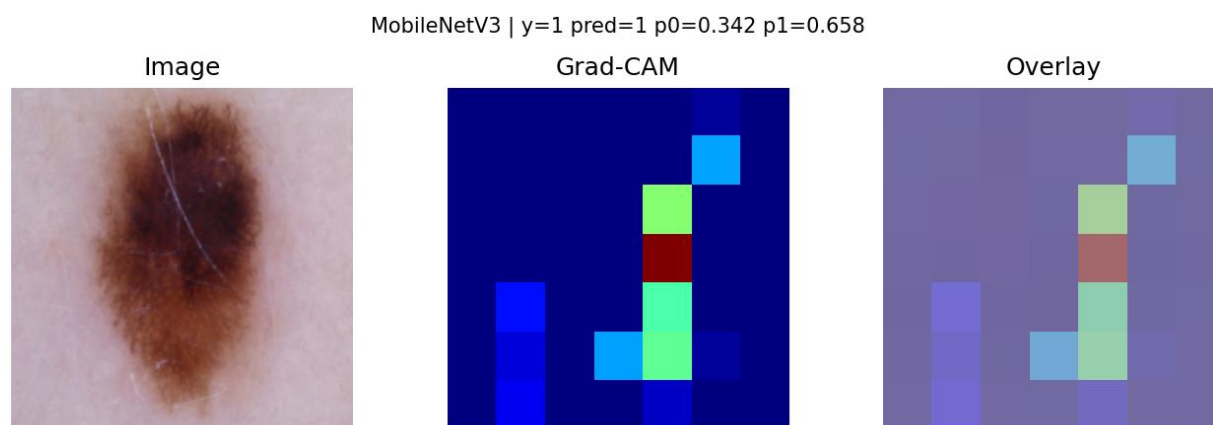


Figure 10: same sample, MobileNet

Grad-CAM Visualization Explanation

The Grad-CAM results illustrate how the two models, MobileNetV3 and ResNet18, focus on different regions of the same lesion image.

- **MobileNetV3:** The activation map is relatively sparse and scattered, highlighting only a small portion of the lesion with moderate confidence ($p1 = 0.658$). This indicates limited localization of important features.
- **ResNet18:** In contrast, the Grad-CAM heatmap is more concentrated and centered on the lesion, showing stronger and more confident activations ($p1 = 0.998$). This reflects better focus on relevant lesion characteristics.

These examples demonstrate that ResNet18 tends to provide clearer and more accurate localization of diagnostic regions compared to MobileNetV3. Similar Grad-CAM patterns and trends were observed across other samples, corroborating this qualitative difference between the models.